Energy Efficiency Competition 2026

RULES

Introduction

The Energy Efficiency Competition, which will take place as part of **the Children and Youth Technology Festival 2026**, is an event organised by the Jelgava Technology Secondary School's technical innovation programme *"IT/Engineering"* (hereinafter, ITC). The competition aims to promote STEAM education and raise awareness of energy-efficient, sustainable mobility among young people through practical, project-based learning. It provides students and young people aged 10 to 25 with an opportunity to learn about mechanics, electronics, and programming in a real engineering environment. It offers students and young people aged 10 to 25 the opportunity to learn about mechanics, electronics, programming, design, and material selection in a real engineering environment by developing, building, and testing energy-efficient vehicles. Competitions develop technical thinking, test result analysis, design optimisation, and problem-solving skills, while also building an understanding of the efficient use of resources. Participants learn to work in a team, plan a project from idea to functioning prototype, and comply with safety and quality requirements. Special attention is paid to accessible and inclusive STEAM education, providing opportunities for participants of different ages in the U12, U16, U25, and Open groups, thereby fostering long-term interest in engineering, technology, and energy-efficient solutions for future mobility. At the same time, the competition aims to ensure a long-term impact by creating open learning resources and a repeatable format that schools and youth organisations across Europe can continue to use in the future. As part of the competition, participants build an electric car and compete live at a venue determined by the organiser.

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1) Time and place of the competition

a) The final competition will take place on 5 and 6 June 2026 at the Zemgale Sports and Recreation Complex (Jelgava Ice Hall), Rīgas iela 11, Jelgava.

b) The competition starts on 6 June at 10:30; the qualification stage starts on 5 June at 12:00.

c) Registration and vehicle compliance with technical requirements (hereinafter referred to as technical inspection) for all groups will take place on 5 June from 10:00 to 18:00. **The Open group will also be inspected on 6 June from 7:00 to 8:00.*

d) The qualification stage for all groups will take place on 5 June from 12:00 to 19:00. **The Open group will also take place on 6 June from 8:00 to 9:30.*

e) Due to unforeseen circumstances, the date of the final competition may be changed, with participants being notified in good time.

f) In case of severe weather conditions, such as heavy rain, wind or other unforeseen circumstances, the organiser may change the programme, suspend it or postpone the races, informing the participants accordingly.

g) Registration and technical inspection can be arranged at another time by prior agreement with the organisers.

2) Participants

a) Age groups:

(I) **U12** – pupils up to and including 5th grade (up to and including 12 years of age);
(II) **U16** – pupils up to and including 9th grade (up to and including 16 years of age);
(III) **U25** – young people and students up to and including 25 years of age;
(IV) **Open group** – an open group of participants with no age restrictions, in which anyone who meets the minimum technical and safety requirements can participate.

b) The open group and U12 age group teams must have at least **two** participants. U16 and U25 teams must have at least **four** participants. The open group, U12 and U16 teams consist of participants and a responsible person (teacher, parent, etc.) who registers the team for the competition.

c) The team must register a driver, a reserve driver, and a team leader.

d) The age group of the team is determined by the oldest member of the team.

e) If the vehicle meets the requirements of the relevant age group, the team is allowed to participate in the older age group. Until the start of the Qualification stage, it is possible to re-register in another age group. **The judges may move the team to the Open group due to technical non-compliance.*

f) At the same time, a team may only participate in one age group in the competition.

g) A participant may not participate in more than two teams.

h) The driver and reserve driver may only be members of one team.

i) A vehicle may not participate in the competition for more than 2 years. **Except for the Open group. See 12.d.*

3) Application and registration

a) Registration for the competition will take place from 1 April 2026 to 15 May 2026 by completing the application form electronically. **The application form will be published on the Facebook website on 1 April.*

b) Applications after the deadline can be made by contacting the organisers and sending the application to the specified e-mail address; such applications will be considered on a case-by-case basis.

c) After receiving the application, the organisers will update the list of registered teams on the website, and after reviewing the application, the status of the application will be indicated on the website.

d) The vehicle must be completed by 1 June. On this date, the team must *check in* by sending a video of the vehicle in motion and additional information (e.g. technical documentation and other materials).

e) The technical inspection takes place by prior appointment or on a first-come, first-served basis. On 5 June from 10:00 to 19:00. **The Open group also on 6 June from 7:00 to 8:00.*

f) If a participant is unable to attend the technical inspection, please contact the organisers in good time.

g) Until 12 May 2026, teams have the opportunity to submit a video about themselves (up to 90 seconds long) to be eligible to choose a number between 1 and 99.

4) Competition stages

a) There are 5 competition stages: the qualification stage, which consists of 4 disciplines: Drag race, Braking efficiency, Reversing and figure driving, Efficiency stage, Technical design solutions stage, Car manufacturing knowledge test stage, and Final stage.

b) Participants are only allowed to enter the competition stages of the track if they successfully pass the technical inspection.

c) In the competition stages, the power limit for the Open group, U12 and U16 is 3 W/kg. **The available power is calculated by multiplying the total weight of the vehicle by the power coefficient. Total weight = vehicle + driver + batteries + other equipment necessary for the competition.*

d) In the competition stages, there is no power limit per kilogram for the U25 group. **There is a technical limitation. See 8.b.*

e) Qualification takes place on 5 June and 6 June before the start of the competition.

f) The efficiency stage will take place on 6 June from 10:30.

g) The technical design solution and automotive knowledge testing stages will take place on 6 June from 10:30 until the final stage.

h) The final stage will take place at the end of the competition, after all previous stages.

4.1) Qualification stage

a) In the qualification stage, there will be at least two attempts for each discipline. **The number of attempts may be changed during the competition, taking into account the wishes of the participants and the competition schedule.*

b) For the qualification stage, total points will be awarded for the overall ranking in four disciplines: drag race, braking efficiency, rollback and figure driving.

c) The overall qualification ranking is determined using a qualification score calculated from the placings in each qualifying discipline. All disciplines have equal weight.

$$K = D + B + A + F \quad (1)$$

, where

K – qualification score;

D – place in drag racing;

B – place in braking efficiency;

A – place in the burnout;

F – place in figure driving;

Participants are ranked in ascending order of K. The participant with the lowest K is awarded 1st place in qualification, the next lowest K is awarded 2nd place, and so on. If two or more participants have the same K, their discipline points are summed and divided by the number of participants with that K.

4.1.1) Drag race

a) After the start signal and visual (flag) signal, the driver starts moving and drives 100 metres in a straight line as fast as possible.

b) The time taken to complete the distance is measured. The time is measured from the start signal to the moment when the front edge of the vehicle crosses the finish line.

c) The maximum speed is recorded.

d) Attempts with a false start are not counted.

e) The best attempt is counted.

4.1.2) Braking efficiency in straight-line motion

a) The driver reaches the maximum possible speed and begins to brake at the mark until the vehicle comes to a complete stop. The braking distance and speed before braking begin are recorded.

b) The maximum speed ± 5% achieved in the drag race must be reached, otherwise both categories will not be evaluated.

c) The best attempt is counted.

d) The average deceleration of the vehicle is evaluated. It can be calculated using the formula:

$$a = \frac{v^2}{2l} \quad (2)$$

, where

a – average deceleration, m/s^2;

v – initial speed, m/s;

l – braking distance, m.

4.1.3) Shuttle run

a) The participant starts at the starting line. After the starting sound and visual (flag) signal, the vehicle driver begins moving and drives to the finish line until the vehicle touches or crosses it. The driver then immediately returns to the start line until the vehicle touches or crosses it. This movement is repeated until the start and finish lines have been reached a total of 7 times.

b) The time of the run is recorded. The time is recorded from the official start signal until the moment when, after reaching the 7th line, the vehicle touches or crosses the finish line.

c) The distance between the start and finish is 20 metres (the total distance to be covered is 140 metres). The track width is 3 metres, and there is also space for turning if necessary. **See Figure 1 in the appendix.*

d) For each obstacle knocked down on the track, a time penalty of 3 seconds shall be added to the participant's recorded time.

e) If the competitor deliberately drives outside the boundaries of the track, the attempt will not be counted.

4.1.4) Manoeuvring Course

a) After the start signal and visual (flag) signal, the participant begins to move and completes the track elements: turns, figure eights and narrowing with a free driving width of 1.5 m. All elements must be completed in full and in the specified order.

b) The time of the run is evaluated. The time is measured from the official start signal to the moment when the vehicle crosses the finish line.

c) The best attempt is counted.

d) For each obstacle knocked down on the track, a time penalty of 3 seconds shall be added to the participant's recorded time.

e) If the vehicle deliberately drives outside the track boundaries, the attempt will not be counted.

4.2) Efficiency stage

a) After the start signal and visual (flag) signal, the participant must complete 2 laps (~1.7 km).

b) At least two attempts are allowed.

c) The best attempt is counted.

d) A minimum average speed must be maintained during the ride:

- U12, Open group – 10 km/h;
- U16 – 15 km/h;
- U25 – 20 km/h.

If the minimum average speed is not reached, the attempt will not be counted. **The minimum average speed may be changed on the day of the competition.*

e) The amount of energy consumed per kilogram of total mass (Wh/kg) in the most efficient ride is evaluated:

$$e = \frac{E}{m} \tag{3}$$

, where

e – specific energy consumption, Wh/kg;

E – amount of energy consumed, Wh;

m – total mass, kg.

f) For information purposes, the distance that a vehicle can travel with 1kWh is calculated:

$$S_{1kWh} = \frac{1000l}{E} \tag{4}$$

, where

S_{1kWh} – Distance that can be travelled with one kWh, km/kWh;

l – distance travelled, km.

4.3) Technical design solution stage

a) The team's work in the vehicle construction process is evaluated. **See Table 1 for evaluation criteria. Appendix.*

4.4) Car manufacturing knowledge test stage

a) Evaluates participants' knowledge of car manufacturing and STEAM.

b) Additional points can be earned by participating in various stands at **the Children and Youth Technology Festival 2026**.

4.5) Final stage

a) The final stage is organised separately for each age group (Open group, U12, U16, U25). Participants in each group start at the same time, lining up one behind the other by the points they earned in the qualifying round, and complete five laps for a total distance of approximately 4.25 km. **If several participants have the same number of points in the qualification, their place is determined by the number of points obtained at the start of the discipline. If this is also equal, participants are lined up randomly or by mutual agreement.*
b) The final place is evaluated.

5) Scoring System

a) The number of points earned in the discipline is calculated from the total number of teams, the value of the discipline and the place earned, rounded to a whole number. The number of points is distributed logarithmically using the following formula:

$$p = N - \frac{(N-1) \times \ln(x)}{\ln(N)} \tag{5}$$

, where

p – number of points;

x – place in the discipline;

N – total number of participants.

**For example, if there are 15 teams, the points are distributed as follows: 15, 12, 10, 8, 7, 6, 5, 5, 4, 4, 3, 3, 2, 2, 1.*

b) Driving disciplines 4.1 – 4.3; The maximum number of points is 70% of all points.
c) In technical design solutions, points are awarded according to the criteria. The maximum number of points is 20% of all points **See Table 1 in the appendix.*
d) In the knowledge test, points are awarded according to the conditions of each station. The maximum number of points is 10% of all points.
e) If the discipline is not completed, no points are awarded.
**In points C and D, the score is a percentage value in order to maintain the importance of the discipline regardless of the number of teams.*

6) Vehicle restrictions and requirements

6.1) General technical and safety requirements

a) All safety aspects must be taken into account during the design, construction and competition to ensure the safety of the driver, participants, volunteers and spectators.
b) Vehicles with one or two wheels are not allowed to participate in the competition.
c) Vehicles that damage the track are prohibited from participating in the competition.
d) Electrical system requirements must be met. **See 6.6.*
e) Brakes must be fitted on at least one axle. Electric regenerative brakes are not included in this requirement.
f) The use of single-piece threaded rods in the suspension and steering system is prohibited.
g) Minimum turning radius ≤ 3 m.

h) The exterior of the vehicle and the driver's surroundings must not have any dangerous sharp protrusions made of dense materials that pose a threat to spectators, participants, the driver and everyone else. **For example, there must be no sharp edges, open profiled pipes, screws, wheel axles, etc.*
i) All electrical wires must be insulated, securely fastened and protected in areas where friction or movement is possible. The frame must not be used as a wire channel.
j) The vehicle must be able to stand independently without external support.
k) Maximum permissible dimensions:
   - width – 1.3 m;
   - height – 1.3 m;
   - length – 3 m.
l) Competition number plates: space must be provided for four numbers – at the front, rear, left and right sides. Dimensions: front/rear – round Ø 20 cm; sides – rectangle 20 × 30 cm (width × height). Numbers are issued by the organiser during registration; teams must provide clear, stable mounting locations that are not obscured by other elements. Numbers must be visible throughout the competition; violations may result in a warning or penalty.
m) The vehicle's propulsion must use electricity. Other energy sources, such as internal combustion engines, are not permitted.
n) The power supply system provided by the organisers must be used. Batteries and the power supply unit must be securely fastened. **The parameters of the power supply system are described in section 8.b.*
o) The vehicle must be capable of reaching a speed of at least ~10 km/h.
p) The maximum speed of the vehicle must not exceed 35 km/h.

6.2) Restrictions and requirements for the Open group

a) Vehicles in the Open group must comply with general technical and safety requirements. **See 6.1.*

6.3) U12 restrictions and requirements

a) Vehicles must be self-built. A vehicle is considered self-built if its frame and functional systems are designed and assembled by the team from individual components, rather than installed as complete, commercially available assemblies or modules. **For example, a complete quad bike front suspension with levers, shock absorbers, brakes and other parts is prohibited. However, combining individual components is permitted. For example: bicycle wheel + scooter brake discs + scooter brake calipers + quad bike shock absorbers – permitted.*
b) Wheel diameter ≥ 152 mm (6 inches).
c) Shopping trolley wheels and any wheels that can rotate freely around a vertical axis are prohibited.
d) The front of the vehicle must have a buffer zone of at least 100 mm from the driver's outstretched legs. Open-type vehicles must have a protective barrier of at least 250 mm vertically, with a maximum deviation of 30° from the vertical. Closed vehicles must have an equivalent frame built into the bodywork.
e) The vehicle must be equipped with a speedometer. If a mobile phone is used, it must be securely fastened.

f) It is also recommended to comply with the requirements for older age groups.

6.4) U16 restrictions and requirements

a) All restrictions and requirements applicable to younger age groups apply to this vehicle, unless supplemented by the requirements listed here.

b) The frame must have a built-in *roll bar* located at least 50 mm above the driver's helmet in the driver's intended position.
c) The steering mechanism must be precise and easy to control. **There must be no free play, and the construction must be rigid.*
d) The vehicle must be equipped with a head restraint.
e) The seat must provide sufficient lateral stability. **There must be lateral support.*
f) The brakes must act on both axles.
g) The vehicle must be equipped with a speedometer; the use of a mobile phone for this purpose is not permitted.
h) The vehicle must be equipped with a solid floor that prevents the driver from touching the ground while driving.
i) The speed regulator must be able to automatically return to the off position when the driver is not holding it. A conventional potentiometer is not permitted.
j) All vehicle electronics must be enclosed; no wires or control unit electronics may be visible. At the same time, all electronics must be easily accessible for technical inspection. **Control units must not be located in hard-to-reach places; connections must be accessible.*
k) The driver must be able to exit the vehicle in less than 10 seconds.
l) A physical copy of the technical documentation must be available for technical inspection, and the technical documentation must be submitted at check-in. **See Appendix 1) Description of technical documentation.*

6.5) U25 restrictions and requirements

a) All restrictions and requirements applicable to the latest groups apply to the vehicle, unless supplemented by the requirements mentioned here.

b) Bicycle brakes, except for hydraulic disc brakes, are prohibited. Brakes on heavier vehicles, such as mopeds, are permitted.
c) There must be one brake system on each axle. The front brakes must be controlled by a foot pedal; the rear brakes must be controlled by a handlebar-mounted lever or a separate pedal.
d) If a hydraulic brake system is used, up to 2 master cylinders are permitted per axle. Brake cylinders that operate a single axle must be combined into a single circuit. If the circuits are identical, a rigid piston connection is possible. **Cables and 3D printed parts must not be used.*
e) Equal braking force must be ensured between the left and right wheels.
f) If two pedals are used, they may be combined after testing. All systems must be able to be activated simultaneously without releasing the steering wheel. Pedal area ≥ 10 cm², width ≥ 3 cm.

g) Each system is tested separately and must be able to stop the vehicle on a 20% incline (11.3°).
h) The vehicle must be equipped with at least a three-point seat belt. Each attachment point must withstand a static load of at least 1 kN.
i) The vehicle must be equipped with rear-view mirrors on each side, with a minimum size of 50 × 50 mm, or with cameras on both sides. If cameras or another alternative system are used, it must provide the driver with uninterrupted rearward visibility and the system must provide a field of vision on each side of at least 70° in the horizontal plane and at least 20° in the vertical plane, covering an area up to 10 m behind the vehicle. The visibility system must provide a clear image in all operating conditions and must be easily accessible for Technical inspection. **The requirements for the field of vision of the visibility system can be seen in Figure 1 in the appendix.*
j) The vehicle must be equipped with an audible warning device (horn) with a sound level of at least 85 dB at a distance of 4 m from the front of the vehicle. It must be possible to activate it without releasing the steering wheel.
k) The vehicle must be equipped with lights: front daytime running lights, front and rear turn signals, rear clearance lights and rear brake lights. An emergency light signal must be included.
l) The vehicle must be equipped with at least one device developed by the team that uses information from the CAN interface in the power supply unit.
m) A fuse or other current interruption device must be installed as close as possible (no further than 300 mm) to the positive terminal of the battery block. Automatically resettable switches are not permitted.
n) The vehicle must be equipped with two clearly marked emergency cut-off switches that disconnect the battery from the system in a single operation. One cut-off switch must be easily accessible from the outside. The other cut-off switch must be easily accessible to the driver.

6.6) Electrical system requirements

a) The voltage of the vehicle's electrical system must not exceed the nominal voltage at any time. **U12, U16 – 42 V; U25, Open group – 52 V.*
b) In order to connect to a secure power supply unit and accounting system, the vehicle's power input must be equipped with an XT60 male connector. **See 8b.*
c) One external battery with a voltage of up to 14.4 V and a capacity of up to 15 Wh is permitted for powering electronics. **U25 auxiliary batteries are not permitted.*
d) Industrial communication or ventilation devices may be powered from internal batteries. Devices must be unmodified and autonomous.
e) The entire drive system must be connected to the output of the unit provided by the organisers. **For group U25, other electronic systems may be connected to the battery before the power supply unit and the scoring system.*
f) Electrical insulation of the vehicle frame or metal structures from all wires is mandatory.

g) All cables must be insulated, mechanically secured and protected from friction or contact with moving parts. All connections must be accessible for visual inspection. Poor-quality temporary connections, such as wire splices, multi-strand wires inserted into terminals, and other uninsulated connections, are not permitted.

h) It is essential to protect the vehicle's electrical system, control unit and battery from dirt, dust, mud and water (including rain). We recommend using the IP44 standard as a guideline.

i) Only the charger specified by the manufacturer or a charging system specifically designed for the battery chemistry in question may be used to charge the battery. Charging may only take place in the location specified by the organiser, in compliance with fire safety requirements.

7) Driver safety

a) Head forwarded driving position is prohibited.

b) The driver must wear a helmet and eye protection on the track.

c) Closed footwear.

d) Long trousers and a jacket/shirt with long sleeves.

e) The use of protective gloves (cycling or motor racing) is recommended.

8) Equipment provided by the organisers

a) Radar for speed measurement.

b) Feeding and scoring system documentation: *https://github.com/Xiorcid/ees-link-examples*.

c) Feeding system ~ 25 pcs.

d) Start/finish systems.

e) Scales for weighing.

f) Fire extinguisher.

g) Tents for teams in the technical area of the competition.

9) Rules at the venue

a) Participants must follow the instructions of the venue staff, organisers and organisers' representatives (track staff, marshals), as well as the internal rules and regulations.

b) While on the track, participants must obey the instructions of the track staff.

c) It is forbidden to deliberately interfere with other participants, provoke dangerous situations, cause deliberate collisions, zigzag, interfere with overtaking or perform other unsportsmanlike actions.

d) The pit stop area must be entered safely and at low speed so that the vehicle can be stopped immediately if necessary. The speed limit is 5 km/h.

e) Drivers must wear a fastened safety helmet on the track; helmets may only be removed at the side of the track, at a safe distance from moving vehicles.

f) When leaving the Pit Stop zone, drivers must ensure that they do not obstruct others; if a collision occurs during departure, the driver performing the departure manoeuvre is responsible.

g) Teams may provide their own symbols with the team name. **Not mandatory.*

h) Between disciplines, the vehicle may be adjusted without replacing parts after a technical inspection; in exceptional cases, the replacement of parts must be agreed with the chief judge. **If a change of parts is planned during the competition, the documentation must describe this possibility, its technical solution and the technical compatibility of all parts with the vehicle. Adjustment of the vehicle without changing parts is permitted without additional conditions.*

i) Flag/signal system:
   - red – race stopped;
   - yellow – dangerous area, overtaking prohibited;
   - green – track clear;
   - black and white – warning of unsportsmanlike behaviour.

j) Evacuation: follow the marshals' instructions; drivers may only leave their vehicles when it is safe to do so.

10) Results and awards

a) Results will be announced within 30 minutes after the end of the last discipline; later, they will be available on the official Facebook page.

b) The awards ceremony will take place at the competition venue after the official results have been announced.

c) The three teams with the highest scores in each group will be awarded prizes. **The judges reserve the right to select the winner according to their own criteria if the scores are equal.*

d) Participants may be awarded special prizes, and the judges reserve the right to create nominations on the day of the competition.

11) Penalties, protests and appeals

a) Types of penalties:

   - warning;
   - time penalty;
   - point deduction;
   - disqualification.

b) Typical violations and penalties (example, specified by the organiser):

   - Number plate not visible – warning / time penalty;
   - Helmet not fastened – warning / disqualification from the race;
   - Dangerous riding / deliberate contact – time penalty / disqualification;
   - Technical non-compliance – prohibited from starting until rectified / disqualification;
   - Unsportsmanlike behaviour by the team – disqualification.

c) Submission of protests: within 15 minutes after the results of the respective discipline have been announced; submitted in writing to the chief judge.

12) Changes to the rules

a) These rules come into force at the time of publication and are binding on all competition participants, their representatives and organisers.

b) By participating in the competition, participants agree to the recording and publication of photo/video material in the press, on the channels of supporters and organisers.

c) Minors require the consent of their legal representative (unless the consent of the school community is provided).

d) In matters not covered by these regulations, decisions shall be made by the competition organisers based on generally accepted competition and safety principles.

e) Changes may be made to the regulations that do not affect the vehicle construction process, dimensions and basic principles, by publishing the changes on the official competition information channel.

f) The organisers welcome and, as far as possible, take into account the recommendations of the participants. Questions about the competition can be addressed to the organisers.

Appendix

Table 1. Points awarded for technical solutions

No	Criterion	Explanation	%, of points obtained in the discipline.
1	Design	Assesses aesthetic appearance, colour coordination, purity of form and overall design quality. The extent to which the design corresponds to functionality is also taken into account.	40
2	Team	Assesses teamwork and involvement during the competition. Communication between participants, division of roles, and the ability to make coordinated decisions within a limited time frame are taken into account, as well as the team's behaviour in unexpected situations. Assesses the clear and structured presentation of the solution to the jury.	30%
3	Additional design or technical solutions	Additional points are awarded for functionality or innovative solutions that are not required but improve the design (e.g. LED indicators, adaptable modules, smart solutions).	30

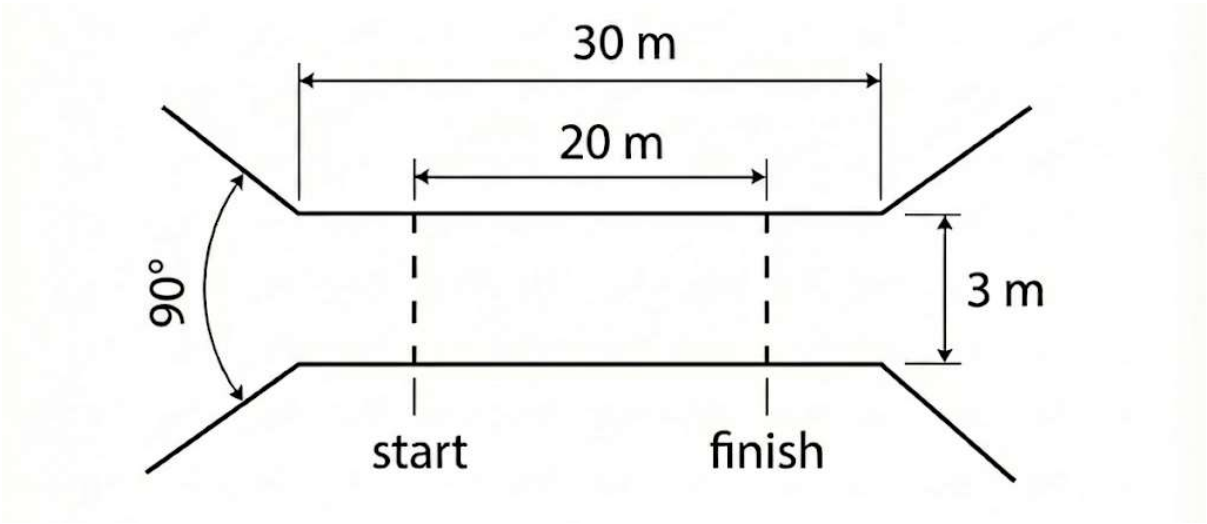


Image1 . Schematic drawing of the track

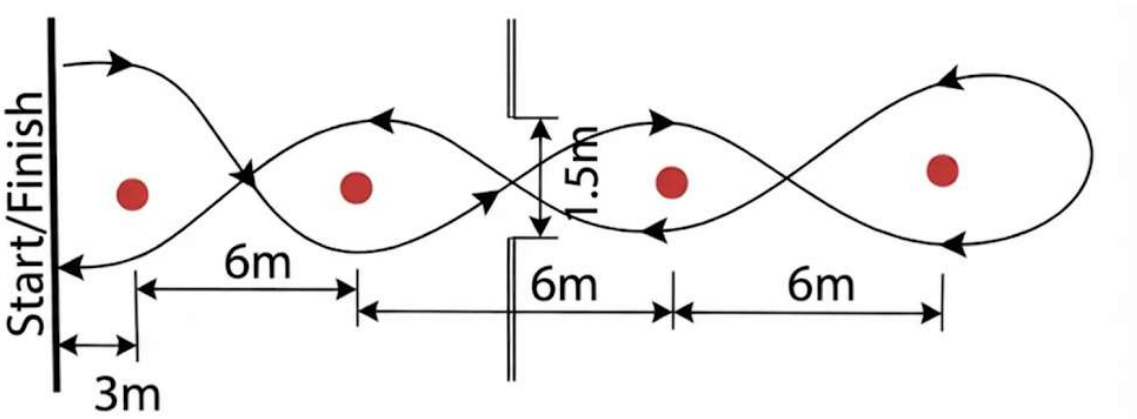


Image2 . Schematic drawing of the figure driving route.

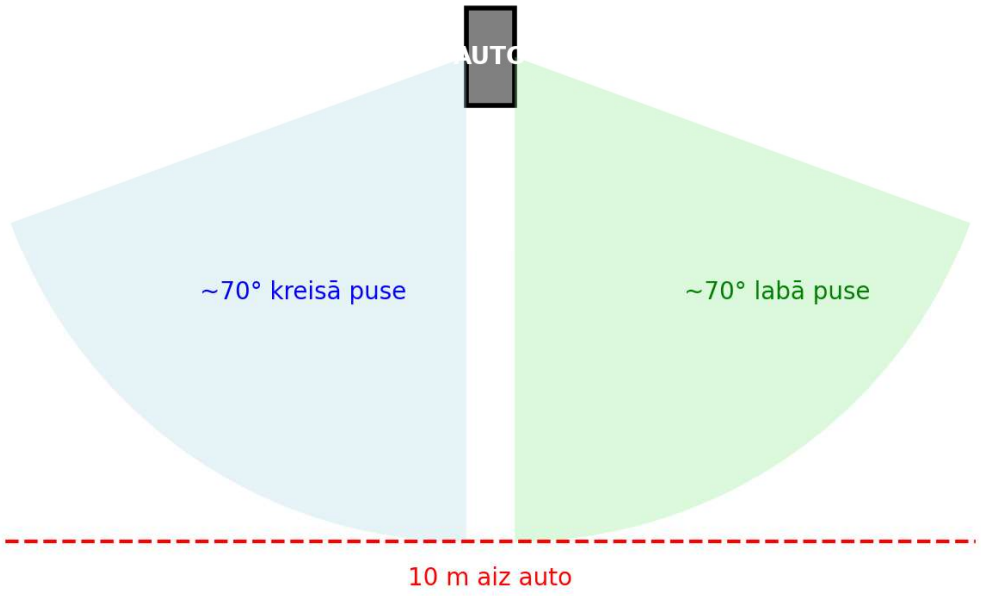


Image3 . Visibility system field of view requirements

1) Description of technical documentation

1.1) Submission deadlines and format

a) Technical documentation must be submitted on 1 June *at check-in*.

b) Changes to the design must be submitted 24 hours before the technical inspection.

c) On the day of the competition, the team must have a printed folder and an identical digital set (USB or cloud link) with them.

d) Drawings must be clearly legible: scale, dimensions, materials.

e) The diagrams must correspond to the actual installation: connector types, wire cross-sections, fuse ratings.

f) File and folder structure

(I) Folders: 01_Mechanics, 02_Electrical, 03_Safety, 04_Test_protocols, 05_BOM_and_Certificates, 06_Procedures.

(II) File names: team_document_vX.Y.pdf (e.g., Vimba_U25_brake_diagram_v1.2.pdf).

Drawings in vector PDF format with scale and dimensions; tables in PDF or XLSX format.

1.2) Minimum requirements for U12, U16 and open group documents

a) Brief description of the vehicle with a photo or sketch. **Not mandatory for the Open group.*

b) Drawing/sketch with main dimensions. **Not mandatory for the open group.*

c) Simple electrical diagram with a list of loads. **Not mandatory for U12 and open groups.*

d) The nominal voltage of the electrical system installed on the power supply unit and powering the drive system must not exceed 42 V (52 V for the Open group) for safety reasons.

1.3) U25 document set (mandatory)

a) Brief description of the vehicle (A4):

- Powertrain, wheelbase, tyres;
- Maximum speed, motor specifications;

b) Drawings: frame, body or monocoque, seat, seat belt mountings with materials and thicknesses, mounting diagram.

c) Electrical system:

- Power supply diagram (battery, motor, control units);
- Fuse list with ratings;
- Control logic block diagram (pedals, brake switches, horn, lighting).

d) Steering and wheel assembly: geometry (Ackermann, mechanical limiters), turning radius justification (≤ 3 m).

e) The nominal voltage of the electrical system installed on the power supply unit and supplying the drive system must not exceed 52 V.

1.4) U25 documentation package (optional)

a) Brake system: hydraulic diagram (master cylinders, pipes, supports), expected braking torque and calculation of how each axle individually can stop the vehicle on a 20% incline.

b) Emergency shutdown and presence control: internal and external E-stop functional diagram (air-gap or equivalent), component layout.

c) Test reports (signed and dated): evacuation test (< 10 s), E-stop test, brake test on a 20% incline (each axle separately and together), horn > 85 dBA at a distance of 4 m

(measurement method), turning radius ≤ 3 m (measurement description), proof of visibility of mirrors or video (real time, brightness in sunlight).

d) Risk assessment and procedures: hazard matrix (electricity, heat, rotating parts), battery charging and storage instructions, pre-start checklist, emergency action plan.